Volume asymptotics for trees of crossing circles

Abstract

We prove two sufficient conditions for the volume asymptotic of the reduced invariant of a fixed ordered tree of crossing circles. One bounds the sum of $d_v/(d_v^2+n_v^2)$, where d_v is degree and n_v is the signed crossing power. The other assigns a decay bound to each directed edge and imposes a finite condition at each vertex. In particular, the latter holds when every vertex has degree at least 2^{28} or absolute crossing power at least 2^{27}. Both conditions permit zero powers at vertices of large degree. The common analysis retains the full finite Newton contraction and its repeated-root derivatives. A joint Fourier calculation proves the summed theorem; a normalized subtree recursion proves the directed theorem without accumulating an exponential loss with depth. Both give a nonzero leading coefficient at every sufficiently large integer level, and the classical primitive-filling theorem identifies the selected action with the volume of the original knot complement. We also prove a centered microscopic meridian limit and, under the summed condition, its explicit first correction.

1 The diagrams and the two sufficient conditions

A crossing circle is an auxiliary component bounding a disk punctured by two main strands. Filling it restores an integer power of the crossing in that disk's original frame. Consider the following literal ordered construction, as in [2, Section 1]. A finite tree $\mathcal{T}=(V,E)$ has at least one edge. Every vertex has its own crossing-circle tangle P_v. A rooted expression is

$$W_v=\left(\prod_{w\in I_v^-}\mathsf{T}(W_w)\right)P_v\left(\prod_{w\in I_v^+}\mathsf{T}(W_w)\right). \tag{1.1}$$

The lists have their specified orders. The turn rotates the whole tangle and its disk frame; it is not just a scalar change of variable. Choose a terminal root and the prescribed cylindrical closure. Keep the actual strand orientations, a reduced cut on a main strand, and zero framing. Require that the endpoint tracing gives one main component and no internal closed component. There are no bare product junctions. This is an input condition on the diagram, not a consequence of the abstract graph.

Write $r=|V|$, $e=|E|=r-1$, and $d_v\geq 1$ for graph degree. At vertex v the physical crossing power is $n_v\in\mathbb{Z}$. The diagram, all powers and orientations are fixed throughout. Only the level $N\geq 3$ tends to infinity. Let $F_N(M)$ be its reduced zero-framed ADO invariant, with unknot value one, in the physical convention of [1]. At $M=1$ it is the reduced Kashaev invariant of the same knot $K_{\mathbf{n}}$. Put

$$q_N=e^{\pi\mathrm{i}/N},\qquad d=N-1,\qquad M=q_N^s,\qquad |s|\leq R<1.$$

For every complex z, the power means $q_N^z=\exp(\pi\mathrm{i}z/N)$; all fractional and complex powers below use this exponential lift. The level degree d is unrelated to d_v.

The first sufficient condition bounds a sum over all vertices:

$$\mathfrak{S}=\sum_{v\in V}\frac{d_v}{d_v^2+n_v^2}\leq 2^{-25}. \tag{1.2}$$

Neither the number of vertices nor the signs or relative sizes of the fixed powers are bounded. A power can be zero when its degree is large enough. The constants in the asymptotic and the initial level may depend on the whole fixed diagram.

Define $J(x) = \int_0^x \log(2\sin \pi u)\,\mathrm{d}u$, $g(x) = J(2x) - 4J(x)$, and $G_0 = g(1/4)$. Near $t = u = 0$ let

$$\psi(0,0) = 2G_0, \qquad \psi_t(t,u) = -\frac{1}{\pi}\operatorname{arsinh}(\tan(t/2)\cos(u/2))\,, \qquad \psi_u(t,u) = \psi_t(u,t). \tag{1.3}$$

Branches continue from zero. The mixed derivatives agree, as checked in Section 3. Set

$$\Psi(\boldsymbol{\tau}) = \sum_{vw\in E} \psi(\tau_v, \tau_w), \qquad F_0 = 2eG_0, \qquad \Phi_{\boldsymbol{\ell}}(\boldsymbol{\tau}) = \Psi(\boldsymbol{\tau}) + \frac{\mathrm{i}}{4\pi}\sum_v n_v\tau_v^2 - \mathrm{i}\sum_v \ell_v\tau_v.$$

Let $\boldsymbol{\tau}_*$ be the small critical point for $\boldsymbol{\ell} = \mathbf{1}$, and $\mathcal{C}(\mathbf{1}) = \Phi_{\mathbf{1}}(\boldsymbol{\tau}_*)$. Existence and uniqueness in the needed chart are proved below.

Theorem 1.1 (Summed degree-and-twist condition)**.** *For the fixed marked knot diagram above, suppose* (1.2)*. There is a constant $D_{\mathcal{T},\mathbf{n}} \neq 0$, independent of the level parity, such that*

$$F_N(1) = \Omega_N N^{3/2} D_{\mathcal{T},\mathbf{n}} e^{N\mathcal{C}(\mathbf{1})}(1 + O(N^{-1})), \qquad \Omega_N = q_N^{d^2(2w(D)+\sum_v n_v)/4}. \tag{1.4}$$

Here $w(D)$ is the oriented writhe of the original diagram and $|\Omega_N| = 1$. The original knot complement is hyperbolic and

$$\lim_{N\to\infty} \frac{2\pi}{N}\log|F_N(1)| = 2\pi\,\mathrm{Re}\,\mathcal{C}(\mathbf{1}) = \mathrm{Vol}(S^3 \setminus K_{\mathbf{n}}). \tag{1.5}$$

Both level parities are included.

For the centered profile put

$$H_N(s) = \frac{\sin \pi s}{N\sin(\pi s/N)}, \qquad A_N(s) = q_N^{(-w(D)-\sum_v n_v)s^2/2}. \tag{1.6}$$

The value at zero is removable. For each edge let $x^*_{vw} = x(\tau_{*,v}, \tau_{*,w})$, where the branch $x(0,0) = 1/4$ is specified by

$$\tan^2(\pi x(t,u)) = \frac{1}{1 + \tan^2(t/2) + \tan^2(u/2)}. \tag{1.7}$$

Theorem 1.2 (First meridian correction under the summed condition)**.** *Under the same hypothesis, $F_N(q_N^s)$ is nonzero on $|s| \le R < 1$ for all sufficiently large N, and uniformly there*

$$\frac{F_N(q_N^s)}{A_N(s)H_N(s)F_N(1)} = 1 - \frac{\pi s^2}{N}\sum_{vw\in E}\cot(\pi x^*_{vw}) + O_R(N^{-2}). \tag{1.8}$$

Consequently the limit in (1.5) *also holds uniformly with $F_N(1)$ replaced by $F_N(q_N^s)$.*

The second sufficient condition evaluates the finite contraction one vertex at a time. Choose any vertex as the last one to be evaluated and direct every edge toward it. At each nonroot vertex choose a number $0 < c_v < 2^{-16}$. This number bounds the decay of the polynomial sent to the parent after division by its value at zero. Section 7 gives explicit inequalities in the degrees, powers and numbers c_v. They require no knowledge of the invariant or its critical value, and they contain no sum over the whole tree.

Theorem 1.3 (Directed degree-and-twist condition)**.** *Suppose the chosen root and numbers c_v satisfy (7.3), (7.8) and (7.9). Then the asymptotic (1.4) and the volume equality (1.5) hold, with a nonzero coefficient independent of level parity. With the same physical scalars as in (1.6), one also has*

$$\frac{F_N(q_N^s)}{A_N(s)H_N(s)F_N(1)} = 1 + O_R(N^{-1}), \qquad |s| \le R < 1. \tag{1.9}$$

These values are nonzero for every sufficiently large N, and (1.5) holds uniformly with $F_N(q_N^s)$ in place of $F_N(1)$.

Corollary 1.4. *Theorem 1.3 holds if, independently at each vertex,*

$$d_v \ge 2^{28} \qquad \text{or} \qquad |n_v| \ge 2^{27}.$$

It also holds under the uniform local condition $\max_v d_v/(d_v^2 + n_v^2) \le 2^{-120}$.

The proof of the corollary is in Section 8 and treats all degrees. A vertex of sufficiently large degree may have zero power. The root used for the directed criterion only orders the finite scalar contractions; it need not be the terminal root used to draw the original knot. No tangle or framing is changed by that choice.

The two criteria have separate proofs. The summed condition gives the explicit correction in Theorem 1.2; the directed proof establishes (1.9) without asserting that sharper expansion. In both cases the tree and all powers are fixed before N tends to infinity. The implied constants and initial levels may depend on that fixed diagram and on R. The curvature bounds in the directed proof do not acquire a factor from the depth of the tree. These are microscopic meridian statements: s is bounded and q_N^s tends to one. No expansion on a fixed neighborhood of $M = 1$ or all-orders annihilator theorem is asserted here.

The exact finite inputs are [1, Proposition 6.1, Theorem 7.1 and Proposition 8.1]: the crossing polynomials and traces, the original tree contraction, and its scalar factors. The canonical factorial of [3, Lemma 2.1] fixes the holomorphic interpolation. Sections 2–5 prove the polynomial estimates, edge asymptotic and exact Fourier trace used by both arguments; they assume neither degree-and-twist condition. Section 6 proves the summed theorem, and Section 7 proves the directed theorem. The classical geometric input is [2, Theorem 1.1]. Each analytic proof checks its threshold and identifies its own critical point with the original primitive filled metric.

2 The original finite contraction and its discarded indices

2.1 Fixed coordinates and the physical factors

Write $\theta = \pi/N$, $\eta = (-1)^{N-1}$, and let $\mathsf{T}_N(z)$ be the monic polynomial with $\mathsf{T}_N(t + t^{-1}) = t^N + t^{-N}$. The pair algebra at a vertex has relation $P_\sigma = \mathsf{T}_N + 2\cos(2\pi\sigma)$, with $\sigma = 0, s$, or $-s$ prescribed by its oriented pair type. The common vector-space basis is

$$e_k(z) = S_k^{-2} \prod_{j<k} (c_j - z), \qquad S_k = \prod_{j=1}^{k} 2\sin(j\theta), \qquad c_j = 2\cos((2j+1)\theta), \quad 0 \le k < N. \tag{2.1}$$

An empty product is one. A canonical representative has degree less than N, obtained using its actual relation P_σ. The different relations are not identified as algebras.

The closing functional and paired edge factors are

$$\phi_\sigma(f) = \eta N^2 [z^d] \operatorname{rem}_{P_\sigma} f, \qquad \phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}, \tag{2.2}$$

$$W_k(s)=\prod_{j=1}^{k}\left(1-\frac{\sin^2(\pi s/N)}{\sin^2(\pi j/N)}\right),\qquad W_kW_{d-k}=H_N^2. \tag{2.3}$$

Every physical turn contributes its prescribed scalar $\gamma_e(s)\in\{1,q_N^{s^2/2},q_N^{-s^2/2}\}$. Lowering its index with (2.2) identifies the equal indices on the two ends of the edge and leaves γ_eH_N/W_k, not its reciprocal.

Let p_v be the actual crossing power, including the carrier specified by the original oriented diagram. The exact identity of [1, Theorem 7.1] is

$$\begin{aligned} Z_N(s)&=\sum_{\boldsymbol{k}\in\{0,\dots,d\}^E}\prod_v\phi_{\sigma_v}\left(p_v\prod_{e\ni v}e_{k_e}\right)\prod_e\gamma_e(s)\frac{H_N(s)}{W_{k_e}(s)},\\ F_N(q_N^s)&=\vartheta_{d+s}^{-w(D)}H_N(s)Z_N(s),\qquad \vartheta_{d+s}=q_N^{(s^2-d^2)/2}. \end{aligned} \tag{2.4}$$

Thus cups, caps, physical turns, negative crossings, the pivotal closure and the one reduced cut are already fixed finite data.

On every disk $R<1$, all factors H_N/W_k and their inverses are bounded independently of N,k. For example split the factors in (2.3) according to $\min(j,N-j)\le J$ or $>J$. The latter logarithms are bounded by a summable constant times $\min(j,N-j)^{-2}$. For fixed J the former factors have no zeros on the disk; their endpoint limits are $1-s^2/j^2$, also zero-free. Compactness, including the finitely many small N, gives the bound and its inverse. The same argument applies to H_N.

2.2 Bounds that retain the first normal

Let $m=\lfloor d/2\rfloor$. Indices $k\le m$ will be called low, the others high. For a canonical polynomial use the complete norm

$$\|f\|_{\mathcal{J}}=\max\{|f(c)|,\ |f'(c)|:\ c\text{ a double node of }P_0;\ |f(-2)|\text{ if }N\text{ is odd}\}.$$

The derivative at the odd simple node is absent. All first derivatives at double nodes are retained.

Lemma 2.1. *For fixed $R<1$, a fixed number of factors and fixed integer crossing powers, quotient multiplication, closing, and those crossing powers have polynomial cost in $\|\cdot\|_{\mathcal{J}}$. Uniformly in all indices,*

$$\|e_k\|_{\mathcal{J}}\le CN^A\begin{cases}\exp(Ng(k/N)), & 0\le k\le m,\\ 1, & m<k<N.\end{cases} \tag{2.5}$$

The constants in these statements do not depend on s in its disk.

Proof. We give the conditioning and endpoint details because ignoring derivatives would make the high-index assertion insufficient. The distinct complete-node polynomial is

$$Q_N(z)=\begin{cases}2T_{N/2}(z/2), & N\text{ even},\\ (z+2)\{U_m(z/2)-U_{m-1}(z/2)\}, & N=2m+1.\end{cases}$$

Here T_j,U_j are the ordinary Chebyshev polynomials. Its interval norm and derivatives are polynomial in N. At its roots, $|Q_N'|\ge N/2$ in the even case and at least $N/2$ in the odd case: for an odd interior root $z=2\cos t$, $|Q_N'(z)|=N/(2\sin(t/2))$, while $|Q_N'(-2)|=N$. The mean-value formula bounds each Lagrange cardinal by a polynomial in N.

For a general polynomial of degree $< N$, interpolate its values by I of degree $< \deg Q_N$. Write $f - I = Q_N h$. The values of h at the double nodes are $(f'(c) - I'(c))/Q_N'(c)$. These determine h, whose degree is less than their number. Use Q_N for this interpolation at even levels and $Q_N/(z+2)$ at odd levels; the same cardinal bound applies. Consequently the interval norm, the complete normal norm, and the Chebyshev coefficient sum are polynomially equivalent. One may bound derivatives without any conditioning assumption: a degree-n Chebyshev expansion has coefficients at most twice its interval norm, and $\|T_j'(z/2)\| \le j^2/2$.

Write $C_j(z) = (t^j + t^{-j})/2$ when $z = t + t^{-1}$. The product identity $2C_iC_j = C_{i+j} + C_{|i-j|}$ makes the coefficient sum submultiplicative. Modulo P_σ,

$$C_N = -\cos(2\pi\sigma), \qquad \operatorname{rem} C_{N+j} = -2\cos(2\pi\sigma)C_j - C_{N-j} \quad (0 < j < N).$$

Iteration bounds reduction of degree at most hN by $(2\cosh(2\pi R) + 1)^h$, for fixed h. Finally $[z^d]C_d = 1/2$. These facts prove the asserted polynomial costs of multiplication and ϕ_σ.

The actual crossing polynomial and its inverse are holomorphic through the collisions in $|\sigma| < 1$, with the normals of [1, Proposition 6.1]. On a parameter circle of radius $R' \in (R, 1)$, $R' \ne 1/2$, Lagrange interpolation has denominator $\sin(2\pi\sigma)$ bounded away from zero. Its nodes have angular imaginary part $O_{R'}(N^{-1})$. The divided differences of T_N and their derivatives on that ellipse are $O_{R'}(N^2)$ and $O_{R'}(N^4)$: differentiate $2\cos(Nt)$, with endpoint limits interpreted by its polynomial expression. The fixed-power eigenvalues, including negative powers, are bounded there. Thus the canonical crossing and its first derivative have polynomial interval norm. Holomorphy and the maximum principle extend the bound through all collisions to the disk. The mixed orders use their actual fixed reference crossing and the bounded scalars of the same proposition, so they satisfy the same conclusion.

For low k, all values $e_k(c_j)$ at distinct nodes are nonnegative and bounded by $e_k(-2)$. Interpolation by Q_N therefore gives $\|e_k\|_{\mathcal{J}} \le N^A e_k(-2)$. Uniform sum–integral comparison of the midpoint cosine product and sine factorial gives

$$\log e_k(-2) = Ng(k/N) + O(\log N), \qquad 0 \le k \le m.$$

This remains uniform at both endpoints: removing the first and last mesh interval leaves a smooth logarithm, while each removed logarithmic interval contributes $O(\log N)$.

For high k, put $l = N - k \le \lfloor N/2 \rfloor$ and $B_j(z) = \prod_{a<j}(c_a - z)$. Reflection $c_{d-a} = c_a$ gives $B_kB_l = (-1)^N P_0$. Thus e_k vanishes at every node. At a double node indexed by $0 \le j < \lfloor N/2 \rfloor$, its derivative is zero unless $j < l$, and otherwise

$$|e_k'(c_j)| = \frac{S_{l-1}^2}{2\sin((2j+1)\theta)S_{l+j}S_{l-1-j}}. \tag{2.6}$$

To obtain this formula, compare quadratic coefficients in $B_kB_l = (-1)^N P_0$, using $P_0''(c_j)/2 = N^2/(4\sin^2((2j+1)\theta))$, $|B_l'(c_j)| = S_{l+j}S_{l-1-j}/(2\sin((2j+1)\theta))$, and $S_kS_{l-1} = N$. The denominator ratio $D_j = S_{l+j}S_{l-1-j}/S_{l-1}^2$ starts at $2\sin(l\theta)$ and is nondecreasing, since

$$\frac{D_{j+1}}{D_j} = \frac{\sin((l+j+1)\theta)}{\sin((l-1-j)\theta)} \ge 1.$$

The two angles sum to at most π. Equation (2.6) is consequently at most $1/(4\sin((2j+1)\theta)\sin(l\theta)) \le N^2/16$. The odd simple-node value is zero. This proves (2.5). □

2.3 Discard before forming an angular sum

Retain $k_e/N \in I = [1/5, 3/10]$ at every edge. Each edge index occurs in two vertex factors of (2.4). Lemma 2.1 therefore bounds the absolute rate of an assignment by $2\sum_{\text{low } e} g(k_e/N)$.

Direct differentiation gives

$$g'(1/4) = 0, \qquad g''(x) = -\frac{4\pi}{\sin 2\pi x} \le -4\pi.$$

Since $g(0) = g(1/2) = 0$, concavity also gives $2G_0 \ge \pi/4$. A low index outside I loses at least $\pi/100$ from F_0; a high index loses at least $2G_0$. The number of assignments is polynomial in N for the fixed tree. We have proved the full-source estimate

$$|\text{all discarded assignments}| \le C_R N^A e^{N(F_0-\eta_0)}, \qquad \eta_0 = \pi/200. \tag{2.7}$$

The exponent loss is independent of tree size and powers; the prefactor and starting level are not. This argument estimates the actual vertex traces, rather than replacing a small spectral norm by a positive coefficient cone.

On the retained indices, the edge sums commute with the polynomial traces. Remove only the already recorded edge scalars and put

$$\begin{aligned} K_N(t,u;s) &= \sum_{k/N\in I} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}, \\ B_N(\boldsymbol{\tau},s) &= \prod_{vw\in E} K_N(\tau_v,\tau_w;s). \end{aligned} \tag{2.8}$$

This is an exact polynomial in each $-2\cos\tau_v$. Its separate degree may be larger than $N-1$; no such bound is assumed in the contours below. The functional ϕ_{σ_v} reduces the whole polynomial in its proper relation.

3 An exact complex edge sum

3.1 The interpolation and its normalization

We use the canonical factorial lemma [3, Lemma 2.1], whose statement we record. On $-1 < \operatorname{Re} z < N$,

$$G_N(z) = \left(\frac{2\pi}{N}\right)^z \Gamma(z+1)\prod_{a\ge 1}\frac{\Gamma(Na+z+1)}{\Gamma(Na-z)(Na)^{2z+1}} \tag{3.1}$$

is holomorphic and nonzero, with positive-base powers. It satisfies $G_N(0) = 1$, $G_N(k) = S_k$, and

$$\frac{G_N(z+1)}{G_N(z)} = 2\sin\frac{\pi(z+1)}{N}, \qquad G_N(z)G_N(d-z) = N, \qquad G_N(s)G_N(-s) = H_N(s)^{-1}.$$

On every compact subset of $0 < \operatorname{Re} x < 1$,

$$\log G_N(Nx) = NJ(x) + \tfrac{1}{2}\operatorname{Log}(2N\sin\pi x) + \frac{\pi\cot\pi x}{12N} + O(N^{-3}). \tag{3.2}$$

All fixed derivatives, and any fixed further inverse-power order, are uniform on smaller compact sets. Logarithms and square roots are continued from the positive real interval. This numbered lemma includes the convergent product and summable Stirling proof; no periodic factor may be inserted into (3.1). For background on those classical expansions see [4, Section 5.11].

The shift equation now proves the exact interpolation needed here. For $u = Nt/(2\pi)$, define

$$\mathcal{B}_N(x,t) = \frac{G_N(d/2+u)G_N(d/2-u)}{G_N(d/2-Nx+u)G_N(d/2-Nx-u)G_N(Nx)^2}. \tag{3.3}$$

At $x = k/N$ this equals $e_k(-2\cos t)$. Indeed successive quotients are the original cosine numerator and sine-factorial denominator of (2.1). The exact edge insertion is

$$\mathcal{I}_N(x,s) = \frac{G_N(Nx)^2}{G_N(Nx-s)G_N(Nx+s)}, \qquad \mathcal{I}_N(k/N,s) = \frac{H_N(s)}{W_k(s)}. \tag{3.4}$$

The second identity follows by iterating the same shift equation and using $G_N(s)G_N(-s) = H_N^{-1}$.

Put

$$b(x,t) = \int_0^x \operatorname{Log}(2\cos(2\pi v) + 2\cos t)\,\mathrm{d}v - 2J(x). \tag{3.5}$$

Branches start at real $x \in I, t = 0$. On a fixed complex tube about I, (3.2) gives

$$\begin{aligned} \mathcal{B}_N(x,t) &= \frac{e^{Nb(x,t)}}{2N\sin\pi x}(1 + N^{-1}b_1(x,t) + O(N^{-2})), \\ \mathcal{I}_N(x,s) &= 1 - \frac{\pi s^2}{N}\cot\pi x + O_R(N^{-2}). \end{aligned} \tag{3.6}$$

The first correction b_1 is independent of s. To check the leading constant, each argument $d/2 \pm u$ has the shift $-1/2$, so its square-root factor is $\sqrt{N}$; the analogous denominator factors cancel these, leaving $(2N\sin\pi x)^{-1}$. Thus no midpoint shift has been discarded. For the second identity use $(\log G_N)''(Nx) = \pi\cot(\pi x)/N + O(N^{-2})$ in the symmetric difference. All remainders are holomorphic with uniform fixed derivatives on a smaller tube.

3.2 The exact stationary phase and an available contour

For $A(x,t,u) = b(x,t) + b(x,u)$, its critical branch satisfies

$$\cos(2\pi x) = \frac{1 - \cos t\cos u}{2 + \cos t + \cos u}, \qquad x(0,0) = 1/4. \tag{3.7}$$

This is equivalent to (1.7). It follows by writing $A_x = 0$, squaring the products of the nonzero factors, and choosing the branch at zero; the original unsquared equation holds there and hence nearby. Substitute $z = \tan\pi v$ into the t-derivative of (3.5). At its critical x, the resulting logarithmic antiderivative is $-\pi^{-1}\operatorname{arsinh}(\tan(t/2)\cos(u/2))$. Thus $\psi = \operatorname{stat}_x A$ is exactly (1.3). In particular ψ is separately even.

We need a domain for the finite sum, not just this limiting phase. On $|t|, |u| \le 1/4$, $|\tan(t/2)| < 0.128$; the tangent form of (3.7) implies

$$|\cos(2\pi x)| < 0.017, \qquad |x(t,u) - 1/4| < 0.003.$$

On $\operatorname{Re} x \in I$, $|\operatorname{Im} x| \le 1/50$, direct differentiation gives

$$A_{xx} = \sum_{\xi=t,u}\left[-\frac{2\pi\sin 2\pi x}{\cos 2\pi x + \cos\xi} - 2\pi\cot\pi x\right], \qquad |A_{xx}| < 44, \quad \operatorname{Re} A_{xx} < -12. \tag{3.8}$$

Here are sufficient elementary bounds: $|\cos 2\pi x| < 0.338$, $|\cos\xi - 1| < 0.032$, $|\sin 2\pi x| < 1.009$, and $\operatorname{Re}\cot\pi x > 0.72$. The real part of $\sin 2\pi x/(\cos 2\pi x + \cos\xi)$ exceeds 0.328: its conjugated numerator is at least $0.95(0.655) - 0.040(0.159)$, and its squared denominator is at most 1.37^2. These also prove the modulus bound.

Enclose precisely the retained integers k by a rectangle in the x-plane with vertical half-grid sides and horizontal height $h = 1/500$. The residue formula with $\pi N\cot(\pi N x)$ equals the finite sum exactly. On a vertical half-grid side the cotangent has modulus at most

one. On horizontal sides it equals $\mp\mathrm{i} + O(e^{-2\pi Nh})$. If $X = \mathrm{Re}(x - x_*), Y = \mathrm{Im}(x - x_*)$, Taylor's formula and (3.8) imply

$$\mathrm{Re}\{A(x) - A(x_*)\} \le -6X^2 + 44|XY| + 22Y^2. \tag{3.9}$$

On a vertical side, for all large N, $|X| \ge 0.047 - o(1)$ and $|Y| \le 0.005$; the right side is below $-0.002 + o(1)$. On a horizontal side its maximum in X is at most $(308/3)Y^2 \le 77/30000$, less than $2\pi h = \pi/250$. All nonconstant cotangent modes and vertical currents therefore have a fixed exponential loss relative to $e^{N\psi}$.

The constant horizontal terms give N times a real integral. Translate this integral to the real tangent through x_*. Its endpoint connectors have the same loss, and $\mathrm{Re}\, A_{xx} < -12$ gives a nondegenerate Gaussian. The ordinary Taylor proof of the saddle expansion applies uniformly: outside a fixed small saddle interval there is a fixed loss; inside it put $x - x_* = N^{-1/2}z$, expand, and integrate the Gaussian-dominated remainder. Odd terms integrate to zero. This also gives any fixed order and its parameter derivatives. All Gamma arguments in (3.3) stay in regular bulk strips throughout these contours.

It follows that

$$\begin{aligned} K_N(t,u;0) &= N^{-3/2}a_e(t,u)e^{N\psi(t,u)}(1 + N^{-1}a_{e,1}(t,u) + O(N^{-2})), \\ a_e &= \frac{\sqrt{2\pi}}{4\sin^2(\pi x_*)\sqrt{-A_{xx}(x_*,t,u)}}. \end{aligned} \tag{3.10}$$

The square root is positive at zero and continued on the disk; no factor vanishes there. In particular $a_e(0,0) = 1/4$. Applying the same contour to (3.4) proves

$$\begin{aligned} B_N(\boldsymbol{\tau},0) &= N^{-3e/2}a(\boldsymbol{\tau})e^{N\Psi(\boldsymbol{\tau})}(1 + N^{-1}a_1(\boldsymbol{\tau}) + O(N^{-2})), \\ \frac{B_N(\boldsymbol{\tau},s)}{B_N(\boldsymbol{\tau},0)} &= 1 - \frac{\pi s^2}{N}\sum_{vw\in E}\cot(\pi x(\tau_v,\tau_w)) + O_R(N^{-2}), \qquad a = \prod_e a_e. \end{aligned} \tag{3.11}$$

For a holomorphic insertion, its first saddle correction is $O(N^{-1})$, since the linear Gaussian term is odd. Multiplication by its prefactor $1/N$ in (3.6) therefore makes the last remainder $O(N^{-2})$. The common a_1 cancels in this ratio.

4 Angular bounds and weighted curvature

4.1 A bound on the entire real angular period

Let $N \ge 160$, $k/N \in I$, and $r_j = \cos((2j+1)\pi/N)$. The exact product is

$$\frac{e_k(-2\cos z)}{e_k(-2)} = \prod_{j<k} Q_{r_j}(z), \qquad Q_r(z) = \frac{r + \cos z}{1 + r}.$$

Reserve the first $m_0 = \lfloor N/32 \rfloor$ factors, where $r_j > \cos(\pi/16) > 9/10$. There are at most $N/20 + 1$ negative parameters, all $> -2/5$. Pair them injectively with parameters among the next $N/20 + 1$ positions, which are $> 7/10$; these positions are disjoint from the negative ones since $m_0 + N/20 + 1 \le N/8$. All remaining unpaired parameters are nonnegative.

For a pair $r > -2/5, s > 7/10$, the coefficients of

$$(z^2 + 2rz + 1)(z^2 + 2sz + 1)$$

are $1, 2(r+s), 2 + 4rs, 2(r+s), 1$, all positive. Normalized at $z = 1$, with its central Laurent power removed, it is a symmetric probability measure on $\{-2, -1, 0, 1, 2\}$. A remaining single

factor has the analogous interpretation on $\{-1,0,1\}$. Its complex-angular modulus is at most $\cosh(by) \le e^{b^2y^2/2}$, where b is its Laurent degree.

For a reserved factor, put $w = \sin^2(t/2)$. The two linear pieces of $|1 - 2w/(1+r)|$ give $|Q_r(t)| \le 1 - w/2$. For $|y| \le 1/4$,

$$|r + \cos(t + \mathrm{i}y)|^2 = (r + \cos t)^2 + 2r\cos t(\cosh y - 1) + \sinh^2 y.$$

Since $2(\cosh y - 1) + \sinh^2 y \le 4y^2$, each reserved factor is bounded by $\exp(-w/4 + 2y^2)$. Use $m_0 \ge N/40$, then multiply the paired and reserved estimates:

$$\left|\frac{e_k(-2\cos(t+\mathrm{i}y))}{e_k(-2)}\right| \le \exp N\left(-\frac{\sin^2(t/2)}{160} + 4y^2\right). \tag{4.1}$$

This pairing precedes absolute values. Individual Laurent factors need not have nonnegative coefficients.

Set $a_v = d_v/(2\pi)$. On $|t_v| \le \pi,\ |y_v| \le 1/4$, $\sin^2(t_v/2) \ge t_v^2/\pi^2$, so (4.1), the bounded edge insertions, and the positive zero-angle sum give

$$|B_N(\mathbf{t} + \mathrm{i}\mathbf{y}, s)| \le C_R N^A \exp N\left\{F_0 - \frac{1}{400}\sum_v a_v t_v^2 + 70\sum_v a_v y_v^2\right\}. \tag{4.2}$$

The stronger incidence constants are $1/(80\pi) > 1/400$ and $8\pi < 70$. Endpoint translations $O_R(N^{-1})$ and derivatives cost only polynomial factors for the fixed tree; one may apply Cauchy's formula on radius $1/N$ disks. This is a bound on the entire angular period, not only near its maximum.

4.2 The local nonlinear terms

With $Q = \sqrt{1 - \sin^2(t/2)\sin^2(u/2)}$, differentiation gives

$$\psi_{tt} = -\frac{\cos(u/2)}{2\pi\cos(t/2)Q}, \qquad \psi_{tu} = \frac{\sin(t/2)\sin(u/2)}{2\pi Q}, \qquad \det\psi'' = \frac{1}{4\pi^2}. \tag{4.3}$$

Consequently

$$\Psi = F_0 - \frac{1}{2}\sum_v a_v\tau_v^2 + \mathcal{E}(\boldsymbol{\tau}).$$

On the polydisk $|\tau_v| \le 1/4$, put $\delta = 1/50,\ \varepsilon = 1/25$. Then

$$\left|\frac{\mathcal{E}_v}{\tau_v}\right| \le \delta a_v, \qquad \sum_w |\mathcal{E}_{vw}| \le \varepsilon a_v, \qquad |\mathcal{E}| \le \frac{\varepsilon}{2}\sum_v a_v|\tau_v|^2. \tag{4.4}$$

The quotient extends at zero by separate evenness. For explicit bounds write $k = \cosh(1/8) - 1 < 0.008$, $\sigma = \sinh(1/8) < 0.126$, and $C = (1-\sigma^4)^{-1/2}$. The normalized diagonal error in (4.3) is at most $2kC/(1-k) + C - 1 < 1/50$, and the off-diagonal bound is $\sigma^2 C$; their sum is $< 1/25$. Summing over incident edges proves the row estimate. Integrate the diagonal error while fixing the other variables to prove the first estimate. For the last, sum the edge quadratic forms and use $2|zw| \le |z|^2 + |w|^2$. No estimate grows with the number of vertices.

5 The exact Fourier trace and its endpoint currents

The identities in this section hold for every fixed finite tree and every signed crossing power. Neither of the two degree-and-twist conditions is used.

For an even trigonometric polynomial $f(t) = f^\sharp(-2\cos t)$, define the exact trace $\mathcal{D}_{n,\sigma,N}[f] = \phi_\sigma(R_\sigma^n f^\sharp)$. There is no restriction on the degree of $f^\sharp$. The actual crossing formula of [1, Proposition 6.1 and equation (6.4)] gives

$$\mathcal{D}_{n,\sigma,N}[f] = \frac{\eta N C_{N,n}(\sigma)}{\sin(2\pi\sigma)} \sum_{j=0}^{N-1} \sin\tau_j \, f(\tau_j) e^{\mathrm{i}nN\tau_j^2/(4\pi)}, \qquad \begin{cases} \tau_j = \dfrac{2\pi}{N}(d/2+\sigma-j), \\ C_{N,n}(\sigma) = q_N^{n(d^2/4-\sigma^2/2)}. \end{cases} \tag{5.1}$$

It is initially written off the colliding grids. Its holomorphic limit is the full trace, including first normals and the odd simple-node value; it is not a distinct-node approximation.

Fix $R < R' < 1$, with $R' \neq 1/2$, and first put the formal parameters σ_v on $|\sigma_v| = R'$. Their denominators are then bounded away from zero. The grid kernel

$$\mathcal{K}_\sigma(t) = \frac{\mathrm{i}N e^{\mathrm{i}Nt}}{e^{\mathrm{i}Nt} - \eta e^{2\pi\mathrm{i}\sigma}}$$

has residue one at each grid point. Integrate it around the rectangle with sides $\pm\pi \pm \mathrm{i}y + 2\pi\sigma/N$. It encloses exactly the N points in (5.1). On its vertical sides,

$$\mathcal{K}_\sigma(\pm\pi + \mathrm{i}v + 2\pi\sigma/N) = \frac{\mathrm{i}N e^{-Nv}}{1+e^{-Nv}}.$$

Thus no argument-dependent pole separation is assumed. The upper and lower horizontal expansions are respectively

$$-\mathrm{i}N \sum_{\ell\geq 1} \eta^\ell e^{-2\pi\mathrm{i}\ell\sigma} e^{\mathrm{i}\ell Nt}, \qquad \mathrm{i}N\left(1 + \sum_{\ell\geq 1} \eta^\ell e^{2\pi\mathrm{i}\ell\sigma} e^{-\mathrm{i}\ell Nt}\right).$$

Each term has an entire integrand, so its translated path can be moved to the fixed horizontal segment, retaining the short endpoint connectors. This is not a deformation of a meromorphic kernel through its moving poles.

The function $\sin t \, f(t) e^{\mathrm{i}nNt^2/(4\pi)}$ is odd. Reflect the upper side by $t \mapsto -t$, with its actual boundary orientation. Combining the two coefficients leaves

$$c_{\ell,N}(\sigma) = \eta^{\ell-1} U_{\ell-1}(\cos 2\pi\sigma), \qquad c_{1,N} = 1. \tag{5.2}$$

The zero-mode integral on the real segment is zero, but moving it there leaves its endpoint currents. The resulting nonzero-charge prefactor is $-N^2 C_{N,n}/(\pi\mathrm{i})$. In particular the reflected charges have already been paired; there will be no later unexamined sum of equal-action signs.

6 The summed degree-and-twist condition

6.1 Critical points with degree in the denominator

Set

$$b_v = \frac{n_v}{2\pi}, \qquad D_v = a_v - \mathrm{i}b_v, \qquad t_0 = \frac{1}{16}, \qquad L_v = \lfloor t_0 |D_v| \rfloor. \tag{6.1}$$

The summed condition implies $\sqrt{d_v^2 + n_v^2} \geq 2^{25/2}\sqrt{d_v}$, so $|D_v| > 32$ and $L_v \geq 1$. This denominator stays large also when $n_v = 0$.

For real continuous charges $0 \leq \ell_v \leq L_v$, the critical equation of $\Phi_{\boldsymbol{\ell}}$ is

$$D_v \tau_v = -\mathrm{i}\ell_v + \mathcal{E}_v(\boldsymbol{\tau}). \tag{6.2}$$

After division by D_v, its right side is a contraction with constant at most ε in the radius-1/4 polydisk, by (4.4) and $a_v \le |D_v|$. Its image has radius at most $t_0 + \delta/4 < 1/4$. It has a unique analytic critical branch $\boldsymbol{\tau}_{\boldsymbol{\ell}}$. Define $\mathcal{C}(\boldsymbol{\ell}) = \Phi_{\boldsymbol{\ell}}(\boldsymbol{\tau}_{\boldsymbol{\ell}})$.

Put $Q_v = \mathcal{E}_v/\tau_v$, with its holomorphic removable extension. The exact equation is

$$\tau_v = -\frac{\mathrm{i}\ell_v}{D_v - Q_v}. \tag{6.3}$$

Since $|Q_v| \le \delta a_v$, define

$$\gamma_+ = \frac{1+\delta}{(1-\delta)^2} = \frac{2550}{2401}, \qquad \gamma_- = \frac{1-\delta}{(1+\delta)^2} = \frac{2450}{2601}.$$

Taking the imaginary part of (6.3) gives

$$\gamma_- \frac{\ell_v a_v}{|D_v|^2} \le -\operatorname{Im}\tau_v \le \gamma_+ \frac{\ell_v a_v}{|D_v|^2}. \tag{6.4}$$

For example the numerator is $\ell_v \operatorname{Re}(D_v - Q_v)$, between $(1-\delta)\ell_v a_v$ and $(1+\delta)\ell_v a_v$; the denominator is bounded by $(1\pm\delta)^2|D_v|^2$. Every positive-charge coordinate therefore has negative imaginary part and is nonzero.

The envelope derivative is $\partial_{\ell_v} \operatorname{Re}\mathcal{C} = \operatorname{Im}\tau_v$. Integrating it one coordinate at a time gives

$$\operatorname{Re}\mathcal{C}(\boldsymbol{\ell}) \le \operatorname{Re}\mathcal{C}(\mathbf{1}) - \frac{\gamma_-}{2}\sum_v \frac{a_v}{|D_v|^2}(\ell_v^2 - 1), \qquad 1 \le \ell_v \le L_v. \tag{6.5}$$

Integrating from $\mathbf{0}$ to $\mathbf{1}$ similarly gives

$$F_0 - \beta \le \operatorname{Re}\mathcal{C}(\mathbf{1}) \le F_0, \qquad \beta = \frac{\gamma_+}{2}\sum_v \frac{a_v}{|D_v|^2} = \pi\gamma_+\mathfrak{S}. \tag{6.6}$$

This compares each charge directly with the filled first-charge action; a comparison only with F_0 would not suffice.

The real part of the Hessian on a real tangent is at most $-(1-\varepsilon)\operatorname{diag}(a_v)$, by the symmetric weighted row bound in (4.4). Its Gaussian determinant is nonzero. Also $|\tau_v| \le t_0/(1-\delta) < \pi/2$, so $\prod_v \sin\tau_v \ne 0$ at every positive-charge saddle.

6.2 Infinite tails with real spectators

Let $A_0 = 1/400$, and use the heights

$$y_v = A_0 t_0 \frac{a_v}{|D_v|}. \tag{6.7}$$

In coordinate v, with all spectators real or on their physical translated grids, (4.2) bounds the exponent on the lower horizontal side by

$$F_0 - A_0 a_v t^2 + |b_v t y_v| - \ell y_v + 70 a_v y_v^2.$$

After maximizing t, the positive quadratic coefficient of y_v^2 is at most

$$70a_v + \frac{100 b_v^2}{a_v} \le \frac{100|D_v|^2}{a_v}.$$

Since $L_v + 1 > t_0|D_v|$, every $\ell \ge L_v + 1$ loses at least

$$\tfrac{1}{2} A_0 t_0^2 a_v \ge \frac{1}{409600\pi}. \tag{6.8}$$

Further charges carry $e^{-N(\ell-L_v-1)y_v}$. The bound $|c_{\ell,N}(\sigma)| \le \ell e^{2\pi R'(\ell-1)}$ is absorbed for sufficiently large N, since this tree and $y_v > 0$ are fixed.

The original vertical sides have loss $\pi^2 a_v/400$. The chirp, finite-charge connector and imaginary Gaussian costs there are at most $\pi A_0 t_0 a_v$, $A_0 t_0^2 a_v$, and $70 A_0^2 t_0^2 a_v$. Their sum is less than half that endpoint loss. The zero-mode current obeys the same estimate. The translated-grid error $O(N^{-1})$ changes only the prefactor.

For completeness, define the finite real-integral functional

$$\mathcal{P}_v[f] = -\frac{N^2 C_{N,n_v}(\sigma_v)}{\pi \mathrm{i}} \sum_{\ell=1}^{L_v} c_{\ell,N}(\sigma_v) \int_{-\pi}^{\pi} \sin t \, f(t) e^{N(\mathrm{i}b_v t^2/2 - \mathrm{i}\ell t)} \, \mathrm{d}t.$$

Use the exact telescoping identity

$$\prod_v \mathcal{D}_v - \prod_v \mathcal{P}_v = \sum_j \left(\prod_{v<j} \mathcal{P}_v \right) (\mathcal{D}_j - \mathcal{P}_j) \left(\prod_{v>j} \mathcal{D}_v \right). \tag{6.9}$$

The left spectators are finite real integrals and the right spectators are physical grids. Each costs only a polynomial in N, with constants depending on fixed powers. Their grid heights are $O(N^{-1})$, not another coordinate's slope-dependent contour height. This is why the loss in (6.8) survives arbitrary relative degrees and twists.

Thus the exact retained contraction is, up to an error with that fixed loss, the finite sum

$$\begin{aligned} &\left(\prod_v -\frac{N^2 C_{N,n_v}(\sigma_v)}{\pi \mathrm{i}} \right) \sum_{1 \le \ell_v \le L_v} \left(\prod_v c_{\ell_v,N}(\sigma_v) \right) \\ &\quad \times \int_{[-\pi,\pi]^r} B_N(\boldsymbol{\tau}, s) \prod_v \sin \tau_v \, e^{N(\mathrm{i}\sum_v b_v \tau_v^2/2 - \mathrm{i}\sum_v \ell_v \tau_v)} \, \mathrm{d}\boldsymbol{\tau}. \end{aligned} \tag{6.10}$$

The *whole difference* is holomorphic in every formal σ_v on the unit disk. The original side has the holomorphic full crossing traces; the finite approximation has polynomial coefficients $c_{\ell,N}$. The maximum principle, one parameter at a time, extends the estimate from the circles to the disks. We can now set $\sigma_v = 0, s, -s$ as the actual types require. This step does not claim that each divided endpoint current separately has a removable singularity.

6.3 The connecting contours and the selected charge

Restrict the real integrals in (6.10) to $[-\rho,\rho]^r$, with $\rho = 1/8$. Their complement loses at least $1/(51200\pi)$ by (4.2); the real chirps and Fourier factors have modulus one. Translate this box to the imaginary part of its own critical point. On a connector write $\tau_v = t_v - \mathrm{i}v_v$, where $0 \le v_v \le -\operatorname{Im}\tau_{\ell,v}$. All these points lie in the radius-1/4 polydisk. Equation (4.4) and Young's inequality give

$$\operatorname{Re}\Phi_{\boldsymbol{\ell}}(\mathbf{t} - \mathrm{i}\mathbf{v}) \le F_0 - \tfrac{1}{8}\sum_v a_v t_v^2 + \sum_v \left[\tfrac{1+\varepsilon}{2} a_v + \frac{2b_v^2}{(3-4\varepsilon)a_v} \right] v_v^2 - \sum_v \ell_v v_v. \tag{6.11}$$

Indeed Young's inequality uses $(3-4\varepsilon)a_v t_v^2/8$ to absorb $b_v t_v v_v$. By (6.4),

$$\gamma_+ \max\{(1+\varepsilon)/2,\ 2/(3-4\varepsilon)\} < 3/4.$$

The last two sums in (6.11) are therefore at most $-\frac{1}{4}\sum_v \ell_v v_v$. Each connector has some $|t_v| = \rho$, and loses at least $1/(1024\pi)$. The negative Fourier terms of all spectators are retained in this calculation; replacing them by independent imaginary growth estimates would lose the result.

On the translated horizontal box the critical point has $|\operatorname{Re}\tau_v| < t_0/(1-\delta) < \rho$, and the real Hessian is strictly negative. Apply the holomorphic packet expansion (3.11) there. Taylor expansion at that critical point gives its Gaussian contribution; the rest has an exponential loss, and the odd Gaussian corrections vanish. For a nonminimal charge the original integral equals this saddle contribution plus an additive error at rate $F_0 - \eta$, where

$$\eta = \frac{1}{819200\pi}. \tag{6.12}$$

We do not assert a relative asymptotic for its original contour if its own action is smaller than that error. Equation (6.5) is what compares its saddle with the first one.

Every discarded source, edge-contour, real-box, tail and connector error has at least the loss (6.12), allowing polynomial prefactors. Finally

$$\beta \le \pi \frac{2550}{2401}\, 2^{-25} < \frac{1}{1638400\pi} = \frac{\eta}{2}. \tag{6.13}$$

All fixed-loss errors are strictly below the first-charge action. Each other retained positive charge is also strictly below it by (6.5). There are finitely many such charges for the fixed tree. These are two distinct comparisons, and both are needed.

6.4 The nonzero coefficient of the whole invariant

Let

$$\mathcal{Q} = -\Psi''(\boldsymbol{\tau}_*) - \mathrm{i}\operatorname{diag}(b_v).$$

Its real quadratic part is positive on real vectors. Choose $\sqrt{\det \mathcal{Q}}$ by the complex Gaussian on that real tangent. This branch continues from positive definite real matrices in the domain with positive real part. Applying (3.11) to the first integral in (6.10) gives the explicit constant

$$D_{\mathcal{T},\mathbf{n}} = \left(-\frac{1}{\pi \mathrm{i}}\right)^r (2\pi)^{r/2} \frac{a(\boldsymbol{\tau}_*) \prod_v \sin \tau_{*,v}}{\sqrt{\det \mathcal{Q}}} \neq 0. \tag{6.14}$$

Each factor has already been proved nonzero. The charge-one coefficient is exactly one at every vertex, even at a colliding or constant grid. Hence no parity-dependent leading cancellation remains.

The powers of N can be read without an asymptotic normalization choice. The r exact trace factors supply N^{2r}. The e finite edge sums supply $N^{-3e/2}$, and the angular Gaussian supplies $N^{-r/2}$. Since $e = r - 1$, the total is $N^{3/2}$. At $s = 0$ all physical turn factors and H_N/W_k are one. Equations (2.4) and (5.1) therefore leave precisely

$$\vartheta_d^{-w(D)} \prod_v C_{N,n_v}(0) = q_N^{d^2(2w(D)+\sum_v n_v)/4} = \Omega_N.$$

Every final turn is included by the original compiler in (2.4); it is not removed in this count. This proves the complete leading term (1.4). All nonminimal charges and fixed-loss errors are absorbed using (6.5) and (6.13). The implied exponential rates and initial levels depend on the fixed tree and powers, as the theorem allows.

6.5 Matching the geometric branch

The analytic action is exactly the action of [2, Theorem 1.1], not just a function with its complete Hessian. Its derivative (1.3) agrees on every edge with that theorem, and the constants agree: $2G_0 = v_{\mathrm{oct}}/\pi$, where v_{oct} is the regular ideal octahedron volume. This last equality also follows directly from the Lobachevsky integral at $\pi/4$.

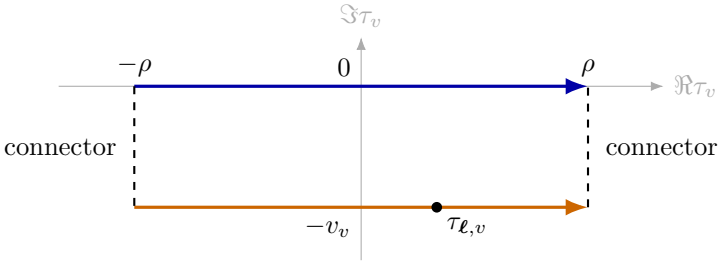


Figure 1: One coordinate of the finite angular deformation. The real box has endpoints $\pm\rho$, with $\rho = 1/8$; its translated horizontal side passes through the critical point. The two vertical sides are retained as connectors, with their boundary orientations when comparing the horizontal integrals. Here $v_v = -\Im\tau_{\boldsymbol{\ell},v} > 0$. Other coordinates can have intermediate downward displacements, so their negative Fourier terms are kept in (6.11). The earlier infinite-charge estimate uses real or translated-grid spectators, before this deformation.

Write $n_v = \epsilon_v - 2p_v$, with the retained parity $\epsilon_v \in \{-1, 0, 1\}$ in each actual disk frame. The geometric theorem fills the original primitive slope $\mu_v + p_v\lambda_v$, with periods

$$U_v = -2\pi\Psi_{\tau_v} - \mathrm{i}\epsilon_v\tau_v, \qquad V_v = 2\mathrm{i}\tau_v.$$

Its unit-charge equation is $\Psi_{\tau_v} + \mathrm{i}n_v\tau_v/(2\pi) = \mathrm{i}$, precisely (6.2) for $\boldsymbol{\ell} = \mathbf{1}$. The summed condition implies $\sqrt{d_v^2 + n_v^2} \geq 2^{25/2} > 630$. Furthermore (6.3) gives

$$|\tau_{*,v}| \leq \frac{2\pi}{(1-\delta)\sqrt{d_v^2 + n_v^2}} < 10^{-2}.$$

Thus our critical point lies in the geometric theorem's uniqueness chart. Its original filled metric is smooth, complete and finite-volume, including zero powers at high-degree vertices; every main cusp stays complete. Its absolute fundamental-chain calculation gives $\mathrm{Vol}(S^3 \setminus K_{\mathbf{n}}) = 2\pi\,\mathrm{Re}\,\mathcal{C}(\mathbf{1})$. There is no appeal here to a long-twist theorem for a zero power, or to a formal exponentiated slope without metric completion. Taking logarithms of the already nonzero (1.4) proves Theorem 1.1.

6.6 The centered main-meridian insertion

The finite scalar ledger of [1, Proposition 8.1] separates exactly $A_N(s)$ from the index-dependent factors H_N/W_k, the shifted crossing grids and the single closing factor $H_N(s)$. In the first-charge coefficient every grid parameter has disappeared because $c_{1,N}(\sigma) = 1$. The chirp has no remaining σ-dependence once its recorded scalar is extracted.

It remains to integrate the second line of (3.11) on the same joint contour. For any fixed holomorphic insertion $h(\boldsymbol{\tau})$, the normalized angular Gaussian evaluates it as $h(\boldsymbol{\tau}_*) + O(N^{-1})$: its linear Gaussian term vanishes, and the quadratic term and its interaction with the cubic phase contribute $O(N^{-1})$. The expansion to the next order and the fixed losses above make this statement uniform on $|s| \leq R$. Apply it to $-\pi s^2 \sum_e \cot(\pi x_e(\boldsymbol{\tau}))$, which already has the factor N^{-1}. The common first saddle correction cancels upon division by $F_N(1)$. The result is (1.8) with a uniform $O_R(N^{-2})$ remainder. In particular the wire values are those at the filled critical point, not the complete values $1/4$.

Both A_N and H_N are nonzero and bounded away from zero on the fixed disk for large N. The ratio in (1.8) tends uniformly to one. This proves uniform eventual nonvanishing and the remaining assertions of Theorem 1.2. No division by an unproved nonzero finite invariant was used to establish the leading term.

7 The directed degree-and-twist condition

The root in this section is an order for evaluating the closed finite contraction. It need not be the terminal root used to draw the original knot. A nonroot vertex sends one polynomial to its parent; the last vertex sends none. The following constants and inequalities reflect that difference.

Put

$$\begin{gathered} a_0 = \frac{1}{2\pi}, \qquad k_0 = 2^{-16}, \qquad R_0 = \frac{1}{8}, \qquad b_0 = 2^{-11}, \\ r_0 = \frac{1}{16}, \qquad t_0 = 2^{-14}, \qquad \chi = 1 - \cos r_0. \end{gathered} \tag{7.1}$$

The parent-variable rectangle is

$$\mathcal{R} = \{u : |\Re u| \le R_0,\ |\Im u| \le b_0\}. \tag{7.2}$$

Choose any vertex as root and direct all edges toward it. For every nonroot vertex choose a number

$$0 < c_v < k_0, \qquad \delta_v = \min\left\{\frac{b_0}{2}, \frac{c_v\chi}{4(1+c_v)}\right\}. \tag{7.3}$$

These are parameters of the criterion. They can differ at every edge. They are not asymptotic quantities to be inferred from the invariant.

Write $q_v = 1$ at a nonroot vertex and $q_v = 0$ at the root. If $\mathrm{Ch}(v)$ is the set of children, define

$$\begin{gathered} a_v = \frac{d_v}{2\pi}, \qquad b_v = \frac{n_v}{2\pi}, \qquad D_v = a_v - ib_v, \\ A_v = \frac{2}{\pi^2}\left(q_v k_0 + \frac{1}{2}\sum_{w\in\mathrm{Ch}(v)} c_w\right), \quad B_v = 16q_v + |\mathrm{Ch}(v)|, \quad C_v = B_v + \frac{b_v^2}{4A_v}. \end{gathered} \tag{7.4}$$

There is at least one incidence at every vertex, so A_v, B_v are positive. Let

$$\delta_v^{\mathrm{in}} = \min\big(\{b_0\} \cup \{\delta_w : w \in \mathrm{Ch}(v)\}\big) \tag{7.5}$$

and set

$$y_v = \min\left\{\frac{\delta_v^{\mathrm{in}}}{2}, \frac{t_0|D_v|}{4C_v}, \frac{A_v\pi^2}{8(\pi|b_v| + t_0|D_v|)}, \sqrt{\frac{A_v\pi^2}{8B_v}}\right\}. \tag{7.6}$$

Finally define two action quantities

$$\beta_v = \frac{210}{361}\frac{a_v}{|D_v|^2}, \qquad \eta_v = \min\left\{\frac{1}{2}t_0|D_v|y_v, \frac{1}{2}A_v\pi^2, A_v r_0^2, \frac{1}{8}a_v r_0^2\right\}. \tag{7.7}$$

The criterion consists of the strict inequalities

$$\boxed{t_0|D_v| \ge 2, \qquad \beta_v < \frac{\eta_v}{2} \quad \text{at every vertex,}} \tag{7.8}$$

and, only at nonroot vertices,

$$\boxed{\beta_v < (k_0 - c_v)\chi.} \tag{7.9}$$

It is a finite elementary test in the degrees, signed powers, root and positive numbers c_v. It uses no value of an unknown quantum invariant and no assumed dominance of a global saddle. Its quantities are local: A_v and the available strip depend on the child budgets, while the exact child scalar will be divided out rather than estimated.

We say that the chosen root and numbers c_v satisfy the directed criterion when (7.3), (7.8) and (7.9) hold. No numerical estimate of an invariant enters this definition. The number β_v bounds the action lost in evaluating just the current trace after exact child constants are divided out. The four entries of η_v pay, respectively, for the infinite charge tail, the original endpoints, localization to a real interval, and its vertical connecting paths. The extra comparison (7.9) is used only to produce an outgoing real bound.

7.1 The full kernel with a fixed parent angle

The basis e_k, trace ϕ_σ, paired factor W_k and edge phase ψ are those of Sections 2–4. The turn retains the scalar γ_e fixed by the original edge orientation; it stays in the exact scalar factors of the contraction. Lemma 2.1 and the complex edge calculation in Section 3 apply before either degree-and-twist condition. Use the exact full kernel

$$K_N(t,u;s)=\sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}. \tag{7.10}$$

The definitions of e_k, W_k and the fixed anti-diagonal trace metric are unchanged. It is a polynomial of degree at most $N-1$ in each of $-2\cos t$ and $-2\cos u$, hence has trigonometric degree at most $N-1$ in either angle. The paired factors and their inverses are bounded for fixed $|s|\le R<1$. Every inequality below permits a polynomial factor in N.

The full-kernel real estimate is

$$|K_N(t,u;s)|\le C_R N^{A_R} e^{N\{2G_0-k_0[(1-\cos t)+(1-\cos u)]\}}, \qquad t,u\in[-\pi,\pi]. \tag{7.11}$$

To verify this, split the sum at $I=[1/5,3/10]$. On I, (4.1) gives angular loss $\sin^2(t/2)/160$ at each end. The positive zero-angle sum has rate $2G_0$. Outside I, the full polynomial bound has loss $\eta_K=\pi/200$, including high-index derivatives. Since $4k_0<\eta_K$ and $2k_0<1/160$, both parts obey (7.11). The bounded paired insertion changes only its polynomial prefactor.

The conditional estimate needed here is

$$|K_N(t+iy,u;s)|\le C_R N^{A_R} e^{N\{\Re\psi(0,u)-k_0(1-\cos t)+16y^2\}}, \quad u\in\mathcal{R}, \quad t\in[-\pi,\pi], \quad |y|\le b_0. \tag{7.12}$$

The parent parameter ranges over the rectangle $\mathcal{R}$. The constant term is the conditional phase $\psi(0,u)$; retaining it is necessary when the exact child constants are divided out.

Here is a proof from the original finite kernel. Split the indices into $I=[1/5,3/10]$ and its complement. The complementary full polynomial, by Lemma 2.1 and (2.7), has the single-edge loss $\eta_K=\pi/200$, including high indices and their complete derivatives. Its ellipse bound is

$$C_R N^{A_R} e^{N\{2G_0-\eta_K+|y|+|\Im u|\}}. \tag{7.13}$$

The edge Hessian in (4.3) gives, on the radius 1/4 bidisk,

$$|\psi_{tt}+a_0|\le a_0/50, \quad |\psi_{uu}+a_0|\le a_0/50, \quad |\psi_{tu}|\le a_0/50, \quad \psi_t(0,u)=\psi_u(t,0)=0. \tag{7.14}$$

In particular $|\psi(0,u)-2G_0|\le |u|^2/10$. Hence (7.13) is smaller than (7.12): the sum $b_0+b_0+(R_0^2+b_0^2)/10+2k_0$ is less than $\eta_K/2$. All these constants are independent of the tree.

For the retained indices write $u = a + ib$. Their factors at a are positive. If $r_j = \cos(2\pi(j+1/2)/N)$, then $r_j > -1/3$ and $r_j + \cos a > 253/384$. Along $|a| \le 1/8$, $|b| \le 2^{-11}$, the denominator $r_j + \cos(a+ib)$ has modulus greater than $3/5$, and

$$\left|\frac{d^2}{du^2}\log(r_j+\cos u)\right| = \left|\frac{-r_j\cos u - 1}{(r_j+\cos u)^2}\right| < 9. \tag{7.15}$$

The first derivative at real a is real. Integrating vertically therefore gives the actual finite-product estimate

$$|e_k(-2\cos(a+ib))| \le e_k(-2\cos a)e^{5Nb^2}, \qquad k/N \in I. \tag{7.16}$$

At the other end retain the original product bound

$$|e_k(-2\cos(t+iy))| \le e^2 e_k(-2)e^{N[-\alpha\sin^2(t/2)+11y^2]}, \qquad \alpha = 1/160 > 1/250. \tag{7.17}$$

The positive retained sum at $(0,a)$ is bounded by a polynomial times $\exp(N\psi(0,a))$, uniformly for real $|a| \le 1/8$. This is the positive one-edge Laplace estimate in Section 3: its unique critical wire is

$$\tan^2(\pi x) = \frac{1}{1+\tan^2(a/2)}, \tag{7.18}$$

which stays in the interior of I. Its second derivative stays negative on this fixed real compact. Equations (7.16)–(7.17), and the second bound in (7.14), consequently give

$$|K_{N,I}(t+iy,u;s)| \le C_R N^{A_R} e^{N\{\Re\psi(0,u)-\alpha\sin^2(t/2)+11y^2+6b^2\}}. \tag{7.19}$$

For $|t| \ge 1/8$,

$$6b_0^2 < \frac{\alpha}{2}\sin^2(1/16), \qquad 2k_0 < \alpha/2. \tag{7.20}$$

For example $\sin(1/16) > 1/(8\pi)$, $\pi^2 < 10$ and $\alpha > 1/250$ already prove the first inequality. This proves (7.12) in this range.

For $|t| \le 1/8$, (3.10) gives the exact nonzero edge asymptotic on the $1/4$ bidisk. Both u and $t+iy$ lie strictly inside that disk. Integrating the first bound of (7.14) gives (7.12) there. The original lattice currents and (7.13) are exponentially smaller. In particular the full kernel, uniformly on this smaller rectangle product, retains

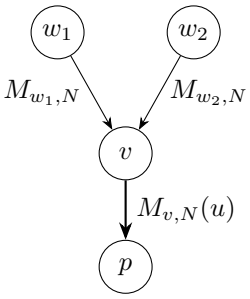
$$K_N(t,u;s) = N^{-3/2}a_e(t,u)e^{N\psi(t,u)}(1+O_R(N^{-1})), \tag{7.21}$$

with the original nonzero holomorphic leading amplitude and midpoint normalization. The outside physical edge gamma is not included in a_e.

7.2 The exact recursion may be rooted anywhere

Lower the fixed Frobenius pairing at every vertex of the original closed source. Every physical turn then identifies equal Newton indices and retains its factor $\gamma_e H_N/W_k$. The resulting finite scalar is a sum over the original edge indices of a product of the original vertex trace tensors. Choosing a root merely specifies the order of these finite contractions. It does not move a cut through a tangle, commute open operators or change a framing.

Consequently the exact recursions of [1, Theorem 7.1] hold for any chosen root:



$$\widehat{M}_{w,N}(t) = M_{w,N}(t)/M_{w,N}(0)$$
$$M_{v,N}(u) = \gamma_{e_v}\left(\prod_w M_{w,N}(0)\right) \cdot \mathcal{D}_v\left[K_N(t,u)\prod_w \widehat{M}_{w,N}(t)\right]$$

At the final root: omit $K_N(t,u)$ and γ_{e_v}. There is no outgoing polynomial.

Figure 2: Edges point toward the vertex whose trace is evaluated next. The exact numbers $M_{w,N}(0)$ are factored out before applying that trace. The outgoing polynomial is then divided by its own exact value at zero. Only a nonroot vertex must supply an outgoing real-decay bound; the final root evaluates a scalar. This diagram describes contraction order, not an isotopy of the knot.

$$M_{v,N}(u) = \gamma_{e_v}(s)\,\mathcal{D}^{(t)}_{n_v,\sigma_v,N}\left[K_N(t,u;s)\prod_{w\in \mathrm{Ch}(v)} M_{w,N}(t)\right] \tag{7.22}$$

at a nonroot vertex, where $\mathcal{D}_{n,\sigma,N}[f] = \phi_\sigma(R^n_\sigma f)$ uses the actual physical crossing polynomial and its Cartan scalar. At the root the recursion is

$$Z_N = \mathcal{D}^{(t)}_{n_o,\sigma_o,N}\left[\prod_{w\in \mathrm{Ch}(o)} M_{w,N}(t)\right] \tag{7.23}$$

at the root. The original global closing, framing and reference factors of [1, Proposition 8.1], and the original single reduced cut H_N, remain outside (7.23). Edge gammas are counted once. The exact normal value $\sigma_v = 0$ and the moving values $\pm s$ retain their actual orientation types.

Each message is an even trigonometric polynomial of degree at most $N-1$ in u, regardless of the subtree. There is no discarded-source error in (7.22)–(7.23).

7.3 The phase class and the variable envelopes

The inductive analytic class is

$$h_v(0) = h_v'(0) = 0, \qquad h_v(-u) = h_v(u), \qquad |h_v''(u) + a_0| \le a_0/20, \quad u \in \mathcal{R}. \tag{7.24}$$

The leading normalized amplitude $\alpha_v(u)/\alpha_v(0)$ is holomorphic and nonzero on a neighborhood of the required smaller compact. Exact subtree constants are retained:

$$\frac{M_{v,N}(u)}{M_{v,N}(0)} = \frac{\alpha_v(u)}{\alpha_v(0)} e^{Nh_v(u)}(1 + O(N^{-1})), \quad u \in \mathcal{R}. \tag{7.25}$$

As usual a slightly smaller rectangle can be used for fixed derivative estimates; the strict margins permit an initially slightly larger one.

The global assertions attached to c_v are

$$\left|\frac{M_{v,N}(t)}{M_{v,N}(0)}\right| \le CN^A e^{-Nc_v(1-\cos t)}, \quad t \in [-\pi,\pi], \tag{7.26}$$

$$\left|\frac{M_{v,N}(t+iy)}{M_{v,N}(0)}\right| \le CN^A e^{N[-(c_v/2)(1-\cos t)+y^2]}, \quad |y| \le \delta_v. \tag{7.27}$$

For canonical representatives of degree less than N, the real sup, Chebyshev and full value-and-first-derivative norms differ by polynomial factors, by Lemma 2.1. A product of degree a fixed multiple of N is first bounded in its real sup norm and then folded in the correct quotient; no equivalence with finite node data is asserted for an unreduced higher-degree polynomial. In particular the reduction and trace retain the repeated-root normals.

Here is the passage from the real bound to the strip. Write the normalized message as a Laurent polynomial $P(e^{it})$, of powers between $-m$ and m, where $m < N$. The subharmonic function $\log|z^m P(z)|$ on the unit disk is bounded by the harmonic function $\log(CN^A) - Nc_v + Nc_v \Re z$. Substitute $z = e^{-y+it}$ and divide by $|z|^m$; for negative y use the reciprocal polynomial. Since $1 - e^{-|y|} \le |y|$, this gives

$$|M_{v,N}(t+iy)/M_{v,N}(0)| \le CN^A e^{N[-c_v(1-\cos t)+(1+c_v)|y|]}. \tag{7.28}$$

For $|t| \ge r_0$ use (7.3). For $|t| < r_0$ use the rectangle expansion and

$$\Re h_v(t+iy) \le -\frac{19a_0}{40}t^2 + \frac{21a_0}{40}y^2. \tag{7.29}$$

Both arguments give (7.27), without a conjectured quadratic estimate for an individual high Newton basis element.

Assume the children satisfy (7.24)–(7.27). At a nonroot vertex set

$$A_v^{\mathrm{loc}}(t,u) = \psi(t,u) + \sum_w h_w(t), \quad C_{v,\ell}(u) = \mathrm{stat}_t\{A_v^{\mathrm{loc}}(t,u) + ib_v t^2/2 - i\ell t\}. \tag{7.30}$$

At the root omit ψ and u. There are d_v incidences in t. Hence

$$A_v^{\mathrm{loc}}(t,u) = \psi(0,u) - a_v t^2/2 + E_v(t,u), \quad |E_{v,tt}|,\ |E_{v,t}/t| \le a_v/20. \tag{7.31}$$

The root constant in this formula is zero. For $0 \le \ell \le L_v = \lfloor t_0|D_v| \rfloor$ the equation $D_v t = -i\ell + E_{v,t}$ contracts in $|t| \le b_0$. Indeed $t_0 + b_0/20 < b_0$. The actual solutions lie in the smaller disk $|t| \le t_0/(19/20)$, leaving room for the translated contours.

Writing $E_{v,t} = tQ_v$, with $|Q_v| \le a_v/20$, gives $t_{v,\ell} = -\mathrm{i}\ell/(D_v - Q_v)$. Its imaginary part is $-\ell\Re(D_v - Q_v)/|D_v - Q_v|^2$. The numerator lies between $19a_v/20$ and $21a_v/20$, and the denominator between $(19|D_v|/20)^2$ and $(21|D_v|/20)^2$. Therefore

$$\frac{380}{441}\frac{\ell a_v}{|D_v|^2} \le -\Im t_{v,\ell} \le \frac{420}{361}\frac{\ell a_v}{|D_v|^2}, \tag{7.32}$$

also for complex parent u. No child phase is asserted to be real on the real axis. Its nonlinear derivative is t times a function bounded by $a_v/20$, so the real part of the denominator remains positive.

The normalized phase $h_v = C_{v,1}(u) - C_{v,1}(0)$ satisfies

$$h_v' = \psi_u(t_{v,1}, u), \qquad h_v'' = \psi_{uu} - \frac{\psi_{ut}^2}{A_{v,tt}^{\mathrm{loc}} + ib_v}. \tag{7.33}$$

Using (7.14) and $|A_{v,tt}^{\mathrm{loc}} + ib_v| \ge (19/20)|D_v| \ge (19/20)a_0$ proves (7.24), since $1/50 + 20/(19 \cdot 2500) < 1/20$. This is uniform on the rectangle. Charge ordering and the current-vertex drop are

$$\Re C_{v,\ell}(u) \le \Re C_{v,1}(u) - \frac{190}{441}\frac{a_v}{|D_v|^2}(\ell^2-1), \quad 0 \le \Re\psi(0,u) - \Re C_{v,1}(u) \le \beta_v. \tag{7.34}$$

At the root again replace $\psi(0,u)$ by zero.

7.4 Actual trace tails with the directed budgets

Divide every child by its exact nonzero value $M_{w,N}(0)$. The induction uses those complex constants exactly; it never estimates them by their complete-source magnitudes. Equations (7.12) and (7.27) imply

$$|\text{normalized trace input}(t+iy,u)| \le CN^A e^{N[\Re\psi(0,u)-A_vt^2+B_vy^2]}, \quad |y| \le \delta_v^{\text{in}}. \tag{7.35}$$

The root constant is zero. The factor $2/\pi^2$ in (7.4) is exactly the inequality $1-\cos t \ge 2t^2/\pi^2$ on the original interval.

Apply the exact trace formula (5.1) and the residue calculation of Section 5 to this normalized input. Here the rectangle has height y_v. The input is an even trigonometric polynomial of degree a fixed multiple of N, so that calculation applies without reduction to a distinct-node model. The paired charge coefficient is (5.2), with first coefficient exactly one. All endpoint currents, including the zero-mode currents, remain to be estimated below.

For $\ell > L_v$, moving the single-grid rectangle by y_v costs, after maximizing the chirp term in the real variable, at most $C_vy_v^2$. Since $\ell > L_v$ implies $\ell > t_0|D_v|$, (7.6) gives a tail loss at least

$$t_0|D_v|y_v - C_vy_v^2 \ge \frac{1}{2}t_0|D_v|y_v. \tag{7.36}$$

Successive charges have the additional factor $\exp(-Ny_v)$. The exact fixed-compact growth of (5.2) is absorbed for all sufficiently large N. This starting level may depend on the fixed vertex.

At an original endpoint $\pm\pi$, the chirp and finite-charge connectors cost at most

$$(\pi|b_v| + t_0|D_v|)y_v + B_vy_v^2 \le A_v\pi^2/4. \tag{7.37}$$

Thus their loss is at least $A_v\pi^2/2$. This includes the original zero-mode current. Nothing is inferred by dropping the endpoint terms of the nonperiodic chirp.

For a retained charge, restrict its real integral to $[-r_0,r_0]$. Its complement loses $A_vr_0^2$ by (7.35). Translate the retained interval to $\Im t_{v,\ell}$. Its endpoints remain in the analytic rectangle. The original local Young inequality, now on this larger real interval, is

$$\begin{aligned}\Re\{A_v^{\text{loc}}(t-iz,u) + ib_v(t-iz)^2/2 - i\ell(t-iz)\} \le \Re\psi(0,u) - a_vt^2/8 \\ + \left[\frac{21}{40}a_v + \frac{2b_v^2}{(14/5)a_v}\right]z^2 - \ell z.\end{aligned} \tag{7.38}$$

For $0 \le z \le -\Im t_{v,\ell}$, the last line after its quadratic term is at most $-\ell z/8$, since $(420/361)\max(21/40, 5/7) = 300/361 < 7/8$. Every such connector therefore loses at least $a_vr_0^2/8$. The translated horizontal saddle has negative real quadratic part at most $-(19/20)a_v$. The physical sine factor is nonzero at $t_{v,\ell}$.

Equations (7.36)–(7.38) give an error of at most $CN^A \exp N[\Re\psi(0,u) - \eta_v]$. Nonminimal charges are their saddle terms plus this absolute error; no relative approximation is asserted

at a much lower nonminimal action. Conditions (7.8) and (7.34) put the whole conditional error strictly below the first charge. Equation (5.2) gives its nonzero coefficient for both parities. This establishes (7.25) and nonvanishing of the exact $M_{v,N}(0)$.

For the child inputs, their exact analytic logarithms differ from the limiting phases by $O(N^{-1})$, with fixed derivatives on the required smaller compact. The finite product of those corrections is retained in the amplitude, or moves the saddle by $O(N^{-1})$. As the tree is fixed, it spends no new exponential budget. The leading amplitude is a product of nonzero edge, child and Gaussian factors. All N-dependent physical crossing and carrier phases remain in $M_{v,N}(0)$; its normalized leading amplitude does not conceal such phases.

The full physical trace and the first-charge approximation are holomorphic in the formal meridian variables. As in the original proof, first take parameter circles avoiding collisions and then apply the maximum principle to their whole difference. This covers every actual $\sigma_v = 0, \pm s$, including normal limits. No isolated divided current is declared removable without its companions.

7.5 Regenerating the outgoing real bound

Lemma 2.1 compares the real sup norm, the Chebyshev norm, and the complete value-and-first-derivative norm by polynomial factors. It also bounds the physical fixed-power crossing and the full quotient trace. These estimates apply to polynomial products of degree a fixed multiple of N; that multiple can depend on the degree of this fixed vertex. For real u, (7.11), the polynomial real sup bounds of the normalized children, the actual fixed-power crossing polynomial, Chebyshev folding in the correct quotient, and the full top-coefficient functional give

$$|M_{v,N}(u)| \le CN^A \left| \prod_w M_{w,N}(0) \right| e^{N[2G_0 - k_0(1-\cos u)]}. \tag{7.39}$$

This retains repeated-root derivatives and all fixed normals. Dividing by the nonzero first-charge value at zero costs at most β_v. For $|u| \ge r_0$, (7.9) absorbs that cost and yields (7.26). For $|u| < r_0$, (7.25) and (7.29) give the stronger estimate, since $c_v < k_0 \ll 19a_0/20$. Equation (7.27) follows by Section 7.3. Thus the induction closes with the selected directed budgets, rather than a common decay constant.

There is no outgoing real bound at the final root. Imposing (7.9) there would be an unnecessary restriction. Its only normalization comparison is the conditional trace estimate (7.8).

7.6 Original action, geometry, and scope

Restore the exact subtree constants successively. Stationary elimination in (7.30) gives the original joint equations

$$\sum_{e \sim v} \psi_{e,\tau_v} + \frac{i n_v \tau_v}{2\pi} = i. \tag{7.40}$$

The exact child constants telescope to the original critical value

$$\mathcal{C}(\mathbf{1}) = \operatorname{stat}_{\boldsymbol{\tau}} \left[\sum_e \psi_e + \frac{i}{4\pi} \sum_v n_v \tau_v^2 - i \sum_v \tau_v \right]. \tag{7.41}$$

Back-substitute from the final root: its first-charge angle belongs to the rectangle, so each child's selected parent value does also, and the same statement continues down the finite tree. Every restored angle has modulus at most $1/[(19/20)|D_v|]$. Thus the vector satisfying (7.40)

lies inside the full simultaneous small-angular uniqueness polydisk, not just the conditional disks used during elimination. The existing contraction identifies it with the same branch as the marked positive charge path of [2, Theorem 1.1]. In fact $20/(19|D_v|) < 10^{-2}$. Indeed (7.8) implies

$$\sqrt{d_v^2 + n_v^2} = 2\pi|D_v| \geq 2\pi\, 2^{15} > 630. \tag{7.42}$$

[2, Theorem 1.1] applies with no sum condition. The edge functions agree: their derivatives are (1.3) and their complete values satisfy $2G_0 = v_{\mathrm{oct}}/\pi$, as in Section 6.4. Write $n_v = \epsilon_v - 2p_v$, with $\epsilon_v \in \{-1, 0, 1\}$, in the original disk frame. The primitive periods supplied by the geometric theorem are

$$U_v = -2\pi\Psi_{\tau_v} - \mathrm{i}\epsilon_v\tau_v, \qquad V_v = 2\mathrm{i}\tau_v.$$

Equation (7.40) therefore gives the full continued relation $U_v + p_v V_v = -2\pi\mathrm{i}$. The imported theorem proves its primitive winding, smooth metric completion, and complete main cusp; we do not infer these facts from an exponentiated equation. Its original ordered two-octahedron cut, odd seam and calibrated action identify $2\pi\Re\mathcal{C}(\mathbf{1})$ with the volume of the actual filled knot complement. This is a geometric input, not a deduction from the quantum saddle alone.

There are $|E|$ edge Gaussians $N^{-3/2}$ and $|V|$ trace Gaussians $N^{3/2}$; $|V| = |E| + 1$ leaves the original $N^{3/2}$ power with nonzero amplitude. The exact first Fourier coefficient is one. The retained edge insertion has leading value one, and the scalar ledger remains precisely (1.6). At $s = 0$, all edge gammas and paired insertions equal one. The exact outside scalar is

$$\Omega_N = \theta_d^{-w(D)} \prod_v C_{N,n_v}(0) = q_N^{d^2(2w(D)+\sum_v n_v)/4}, \qquad \theta_d = q_N^{-d^2/2}.$$

The remaining leading coefficient is independent of N, nonzero, and the same through both parities. This proves the full leading asymptotic in Theorem 1.3. For bounded s, the first-charge coefficient is still one, the paired insertion has leading value one, and all normalized leading amplitudes are unchanged. Restoring the exact scalar ledger of [1, Proposition 8.1] gives $A_N(s)H_N(s)$. The finite product of relative $1 + O_R(N^{-1})$ factors proves (1.9). Its nonvanishing follows uniformly on every fixed disk $R < 1$, since A_N and H_N have no zeros there. Together with the geometric identification above this proves Theorem 1.3.

The criterion allows different strip and real-decay budgets and a different choice of final vertex. It is not the assertion that arbitrary capped short-twist islands can be substituted as black-box messages. Such an island requires a multi-boundary version of the conditional normalization proved here for an actual edge. The one-edge estimates proved here do not supply that extension.

8 Consequences and examples

8.1 A star under the summed condition

In a star with d terminal vertices, central power zero and terminal powers n, the sum in (1.2) is

$$\mathfrak{S} = \frac{1}{d} + \frac{d}{1+n^2}.$$

Taking $d \geq 2^{27}$ and $|n| \geq \sqrt{2^{27}d}$ gives $\mathfrak{S} \leq 2^{-26}$. The diagram must still satisfy the endpoint condition of Section 1; the numerical inequality alone does not give that condition. This example uses only Theorems 1.1 and 1.2.

8.2 A uniform degree-or-twist alternative

The directed criterion has the following simple consequence:

$$|n_v| \geq 2^{27} \quad \text{at every vertex} \quad \Longrightarrow \quad \text{Theorem 1.3.} \tag{8.1}$$

This sufficient constant is independent of the number of vertices, their degrees and the choice of root. Here is a proof covering all degrees simultaneously. Choose $c_v = k_0/2$ at every nonroot vertex and write δ for their common value in (7.3). The elementary cosine estimate gives

$$\chi \geq \frac{r_0^2}{2} - \frac{r_0^4}{24} = 2^{-9}\frac{3071}{3072} > 2^{-10}, \qquad \delta > 2^{-30}. \tag{8.2}$$

At any vertex, including the root, put $\mathcal{R}_v = \sqrt{d_v^2 + n_v^2}$. Then

$$A_v \geq \frac{k_0 d_v}{2\pi^2}, \qquad B_v \leq 16 d_v, \qquad C_v \leq \frac{\mathcal{R}_v^2}{8k_0 d_v}. \tag{8.3}$$

For the last inequality use $16 \leq 1/(8k_0)$. Each of the four branches of the minimum in (7.6) consequently has the following lower bound, respectively:

$$\frac{\delta}{2}, \qquad \frac{t_0 k_0 d_v}{\pi \mathcal{R}_v}, \qquad \frac{k_0 d_v}{16\mathcal{R}_v}, \qquad 2^{-12}. \tag{8.4}$$

The second displayed quantity is smaller than the third and fourth. For the third bound, use $\pi|b_v| + t_0|D_v| \leq \mathcal{R}_v(1 + t_0/\pi)/2$. Thus every branch of (7.7), not just its tail branch, is bounded below by

$$\eta_v \geq \min\left\{\frac{t_0 \mathcal{R}_v \delta}{8\pi}, \frac{t_0^2 k_0 d_v}{4\pi^2}\right\}. \tag{8.5}$$

Indeed the second entry here is smaller than each of the other three entries in (7.7), by their explicit values and $t_0 \ll r_0$. On the other hand

$$\beta_v = \frac{420\pi}{361}\frac{d_v}{\mathcal{R}_v^2} < \frac{4d_v}{\mathcal{R}_v^2}. \tag{8.6}$$

For comparison with half the first entry of (8.5), it suffices that $\mathcal{R}_v^2 > 64\pi/(t_0\delta)$, since $d_v \leq \mathcal{R}_v$. The right side is less than 2^{52}, using $\pi < 4$ and (8.2). For comparison with half the second entry it suffices that $\mathcal{R}_v^2 > 32\pi^2/(t_0^2 k_0)$, whose right side is less than 2^{53}, since $\pi^2 < 10$. Both inequalities follow strictly from $\mathcal{R}_v^2 \geq n_v^2 \geq 2^{54}$. The charge-cut condition in (7.8) follows as well.

For the outgoing patch, the sharper elementary estimate is

$$\beta_v \leq \frac{210\pi}{361|n_v|} \leq \frac{330}{361}\,2^{-26} < \frac{3071}{3072}\,2^{-26} \leq \frac{k_0}{2}\chi. \tag{8.7}$$

Here $\pi < 22/7$, $d_v/(d_v^2 + n_v^2) \leq 1/(2|n_v|)$, and the displayed rational comparison prove (7.9) for every degree. This completes the corollary without a numerical scan or a threshold depending on the tree.

The same proof gives a mixed condition, chosen independently at each vertex:

$$\boxed{\quad d_v \geq 2^{28} \quad \text{or} \quad |n_v| \geq 2^{27} \qquad \text{for every } v. \quad} \tag{8.8}$$

Under the degree alternative, $\mathcal{R}_v^2 \geq d_v^2 \geq 2^{56}$ proves the same two tail comparisons, while

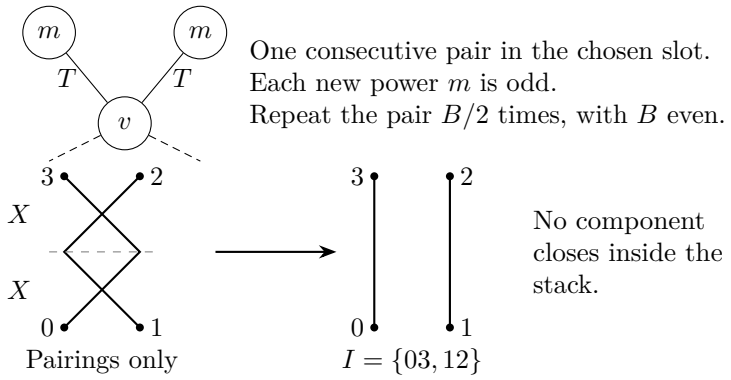


Figure 3: The upper drawing records tangle operations, not a knot projection. In the construction, each drawn pair represents $B/2$ consecutive pairs inserted in one chosen product slot. The lower pairing calculation is $X \circ X = I$, with no internal component; it preserves the original closure's component count, not its knot type.

$$\beta_v \le \frac{420\pi}{361 d_v} \le \frac{330}{361}\, 2^{-26} < \frac{k_0}{2}\chi \tag{8.9}$$

is the same outgoing patch bound. Neither alternative needs to hold uniformly over all vertices. In particular, the high-degree vertices may retain any fixed signed powers, including zero and small nonzero powers. No sum over the number of such vertices occurs.

Here is a precise extension construction illustrating that freedom. Start with any fixed ordered circle-tree core whose original endpoint pairing closes to a knot and has no internal closed component. Retain all its fixed signed powers. At every core vertex choose one of its actual product slots next to its crossing box and insert a consecutive stack of an even number $B \ge 2^{28}$ of new terminal pieces. Every new leaf has a fixed odd power satisfying

$$|n_{\mathrm{leaf}}| \ge 2^{27} + 1. \tag{8.10}$$

The physical quarter-turn of an odd two-strand braid has diagonal pairing $X = \{02, 13\}$; this pairing is invariant under the quarter-turn. An even ordered stack of these leaves has the through pairing $I = \{03, 12\}$ and creates no internal circle. Inserting it in the specified slot therefore preserves the original local endpoint pairing in that actual frame. This is a pairing calculation, not an isotopy removing the new tangles. At the original closing box the same insertion takes place inside its local ball. An added terminal leaf may then be chosen as the recursive root using the original closed-collar move with its induced frame reversal and factor order retained.

Thus the resulting diagram is again a knot, while its knot type may change. Each old vertex now satisfies the degree alternative and each new leaf satisfies the twist alternative in (8.8). The resulting actual knot complement is hyperbolic and satisfies Theorem 1.3. The number and arrangement of prescribed short-power vertices are arbitrary but fixed; the number B does not depend on their total count.

A small uniform local ratio is also sufficient:

Corollary 8.1. *If* $\max_v d_v/(d_v^2 + n_v^2) \le 2^{-120}$*, then the conclusions of Theorem 1.3 hold.*

Proof. If both $d_v < 2^{28}$ and $|n_v| < 2^{27}$, then $d_v^2 + n_v^2 < 2^{57}$. Since $d_v \ge 1$, this would give $d_v/(d_v^2 + n_v^2) > 2^{-57} > 2^{-120}$. Thus every vertex satisfies (8.8). This proof uses the directed theorem, not a summability hypothesis over vertices. □

8.3 A star separating the directed criterion from the simpler tests

Consider an ordered star rooted at its central vertex, of odd degree $d = 30{,}000{,}001$. Set its power to 2 and every terminal power to the fixed odd integer $2^{60} + 1$. Choose $c_v = k_0/2$ on every leaf.

The knot pairing can be checked in the original terminal-root syntax, even though the finite contraction is evaluated from the center. Label the boundary points lower left, lower right, upper right, upper left by $0, 1, 2, 3$. The even and odd braid pairings are respectively

$$I = \{03, 12\}, \qquad X = \{02, 13\}; \qquad T(I) = H = \{01, 23\}, \quad T(X) = X. \tag{8.11}$$

At a terminal root there are $d - 1$ other leaves. Their turned odd braids have pairing X; their ordered product has pairing I because $d - 1$ is even. The central power 2 preserves it. Its actual quarter-turn has pairing H, and stacking the root's odd braid against H still gives H, with no internal closed arc. Cylindrical closure joins 03 and 12, so it closes H into one component. All intermediate products before that last stack are two through-strand braids. There is no hidden internal component.

The criterion can be verified by rational enclosures without expanding this large tree. All leaves have the same data. The center has approximately

$$\beta_o = 1.219\,10^{-7}, \qquad \eta_o = 2.713\,10^{-7}, \qquad \frac{\beta_o}{\eta_o/2} < 0.9. \tag{8.12}$$

Every leaf has a much smaller ratio. The root does not satisfy the nonroot patch inequality for any $c > 0$, because $\beta_o > k_0\chi$. Nevertheless its exact final trace satisfies (7.8). Thus that outgoing patch must not be charged to a root that sends no message.

Moreover

$$\sum_v \frac{d_v}{d_v^2 + n_v^2} > \frac{30{,}000{,}001}{30{,}000{,}001^2 + 4} > 2^{-25}, \tag{8.13}$$

and its maximum is far above 2^{-120}. The example therefore lies beyond the summed condition and the uniform local ratio condition. Its size and all its powers are fixed; it is not an N-growing family. The constant sizes are coarse, and no optimality assertion is intended.

Here are rational enclosures proving the inequalities rather than inferring them from those decimal values. Put

$$\pi_- = 3, \quad \pi_+ = 22/7, \quad c = k_0/2, \quad \chi_- = r_0^2/2 - r_0^4/24, \quad \delta_- = \frac{c\chi_-}{4(1+c)}. \tag{8.14}$$

At the central vertex use the conservative height $y_* = \delta_-/2$. It is at most each branch of (7.6). For example, with

$$D_- = \frac{d}{2\pi_+}, \quad D_+ = \frac{d+1}{2\pi_-}, \quad A_- = \frac{dc}{\pi_+^2}, \quad C_+ = d + \frac{\pi_-^{-2}}{4A_-}, \tag{8.15}$$

the three non-strip comparisons are

$$y_* < \frac{t_0 D_-}{4C_+}, \qquad y_* < \frac{dc}{8(1 + t_0 D_+)}, \qquad y_*^2 < \frac{c}{8}. \tag{8.16}$$

The strip comparison follows from $\delta \geq \delta_-$. Every quantity in (8.14)–(8.16) is rational. Direct cross-multiplication also gives

$$\frac{2[420\pi_+ d/(361(d^2+4))]}{t_0 D_- y_*/2} < \frac{9}{10}, \qquad 2\frac{420\pi_+ d}{361(d^2+4)} < \min\left\{\frac{dc}{2},\ A_- r_0^2,\ \frac{D_- r_0^2}{8}\right\}. \tag{8.17}$$

These compare the upper bound for twice the first-charge drop with all four lower bounds for η_o. The charge cutoff is greater than two.

For a leaf with $n = 2^{60}+1$, take $y_* = t_0 k_0/(2n)$. The same direct rational comparisons, using $D_- = n/(2\pi_+)$, $D_+ = (n+1)/(2\pi_-)$, $A_- = 2k_0/\pi_+^2$, and $C_+ = 16 + [n/(2\pi_-)]^2/(4A_-)$, put this height below each branch of (7.6). They give a lower bound greater than 2^{-49} for every entry of (7.7), while $\beta_v < 2^{-117}$. The outgoing patch follows from $\beta_v < (k_0/2)\chi_-$. This proves the leaf inequalities.

9 Finite reproduction and the scope of the estimates

The exact finite identities and numerical budgets have a combined reproduction command. From the repository root, run

```
python3 -I papers/circle-tree-volume/check.py \
    --output /tmp/circle-tree-volume-check
```

The runner delegates to the two retained suites, in `papers/circle-tree-asymptotics/check.py` and `papers/directed-tree-asymptotics/check.py`. It records both results together with the sources of this paper; the implementations of the checks are not copied into a third suite. The first script checks four symbolic edge identities, finite complex products, and the summed-condition budgets. Its rational enclosures use convergent series for π, sinh and cosh, and its negative controls reject altered budgets. The second checks the directed rational inequalities, endpoint pairings, and finite recursions against full root-grid evaluations with first normals. The recursion comparisons use formal independent vertex parameters; they test the finite algebra and do not establish the existence of additional knot diagrams. Its negative controls include an incorrect outgoing condition at the final root and an odd stack of added leaves. Both records retain the hashes of the sources they check.

These finite calculations do not prove the uniform complex estimates, the Fourier-tail bounds, or the geometric identification. The analytic arguments are written in Sections 3–7.6; the geometric input and its quantitative hypotheses are stated where each theorem uses them. Background for the residue and Gaussian calculations can be found in [4, Sections 2.4(iii)–(iv)]. The contours, endpoint terms and strict losses needed for the finite invariant are specified in the proofs above.

The results concern fixed literal ordered circle trees. In particular, they require the endpoint condition of Section 1, and do not apply solely from the abstract graph. The degree-and-twist conditions do not address a tree that grows with N. Neither proof extends a short-twist subtree to an arbitrary multi-boundary tangle: that would require additional conditional estimates for its boundary variables. These are separate questions from the two volume theorems proved here.

References

[1] *Exact Newton calculus for tangles with a moving meridian*, companion paper, Proposition 6.1, Theorem 7.1 and Proposition 8.1. Repository edition.

[2] *Positive octahedra and primitive fillings for trees of crossing circles*, companion paper, Theorem 1.1. Repository edition.

[3] *Holomorphic source asymptotics and volume limits for strict Conway compositions*, companion paper, Lemma 2.1. Repository edition.

[4] NIST Digital Library of Mathematical Functions, Section 5.11, Gamma-function asymptotics, and Section 2.4, Contour integrals.